LETTERS

Electromagnetically induced transparency with single atoms in a cavity

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Optical nonlinearities offer unique possibilities for the control of light with light. A prominent example is electromagnetically induced transparency (EIT), where the transmission of a probe beam through an optically dense medium is manipulated by means of a control beam[1–3]. Scaling such experiments into the quantum domain with one (or just a few) particles of light and matter will allow for the implementation of quantum computing protocols with atoms and photons[4–7], or the realization of strongly interacting photon gases exhibiting quantum phase transitions of light[8,9]. Reaching these aims is challenging and requires an enhanced matter–light interaction, as provided by cavity quantum electrodynamics[10–12]. Here we demonstrate EIT with a single atom quasi-permanently trapped inside a high-finesse optical cavity. The atom acts as a quantum-optical transistor with the ability to coherently control[13] the transmission of light through the cavity. We investigate the scaling of EIT when the atom number is increased one-by-one. The measured spectra are in excellent agreement with a theoretical model. Merging EIT with cavity quantum electrodynamics and single quanta of matter is likely to become the cornerstone for novel applications, such as dynamic control of the photon statistics of propagating light fields[14] or the engineering of Fock state superpositions of flying light pulses[15].

Remarkable progress has been achieved towards the optical manipulation of light by means of single quantum emitters. First realizations of optical transistors operating on tightly focused laser beams have been demonstrated with individual molecules, quantum dots or nitrogen vacancies embedded in suitable host materials[13,16–18]. However, the weak light–matter coupling that has been reached in these experiments limits the control capabilities considerably. Moreover, increasing the number of particles is not a straightforward task, owing to the difficulties in the preparation of identical quantum radiators.

With this backdrop, a promising avenue is to trap a register of atoms inside an optical cavity[19,20]. High-reflectivity mirrors increase the optical path length and can amplify the matter–light interaction into the strong coupling regime. Optical control has already been achieved in single-atom experiments, including the production of single photons with controlled waveform[21–23], the generation of polarization-entangled photon pairs[24], and the state transfer between a faint laser pulse and a single atom[25]. Incorporating EIT will boost the capabilities of cavity quantum electrodynamics (QED) from the production of single photons towards the coherent manipulation of propagating quantum light fields. For a system with many individually addressable atoms, this will ultimately lead to the realization of a quantum network[26], where the generation, propagation and absorption of light are coherently controlled at the quantum level.

In this work, we coherently control the optical properties of a coupled atom–cavity system through the use of EIT. The heart of the apparatus consists of a high-finesse optical cavity with mirrors separated by 495 μm, a TEM_{00}-mode waist of 30 μm and a finesse of 56,000. The cavity operates in the intermediate coupling regime with $(g_0, \kappa, \gamma) = 2\pi \times (4.5, 2.9, 3.0)$ MHz, where g_0 denotes the atom–cavity coupling constant at a field antinode for the ^{87}Rb $5S_{1/2}$ $F = 1 \leftrightarrow 5P_{3/2}$ $F' = 1$ transition at 780 nm, κ is the cavity field decay rate and γ the atomic polarization decay rate. The atoms are trapped inside the cavity in a far-detuned standing-wave dipole trap, resulting in an average a.c.-Stark shift of 5 MHz. The cavity is stabilized to the $F = 1 \leftrightarrow F' = 1$ transition via a reference laser ($\lambda = 785$ nm). This leads to a bare atom–cavity detuning of 2 MHz. Light scattered during cooling intervals is used to obtain images of the trapped atoms with a CCD camera. This allows the precise determination of the number and position of atoms inside the cavity mode during a given experimental run.

In order to demonstrate EIT in the regime of single atoms, we record transmission spectra of the atom–cavity system under three distinct physical conditions. The transmission is measured with a weak probe laser near-resonant with the $F = 1 \leftrightarrow F' = 1$ transition applied along the cavity axis. In the first step of our experimental protocol, we optically shelve the atom in the hyperfine state $F = 2$, therefore effectively decoupling it from the cavity (Fig. 1a). This yields an empty-cavity transmission spectrum used as a reference. In the second step, the atom is prepared in $F = 1$, such that we realize the case of a two-level atom coupled to the cavity (Fig. 1b). In the third step, we apply an additional control laser transverse to the cavity axis and resonant with the $F = 2 \leftrightarrow F' = 1$ transition; this is named the 'cavity EIT' configuration (Fig. 1c). This forms a Λ-level scheme suitable for the generation of a coherent dark state. The experimental protocol is continuously repeated at a 25-Hz rate while the probe laser frequency is shifted for every repetition cycle. Thus, we simultaneously measure the three transmission spectra for a given number of trapped atoms.

We introduce the main features of cavity EIT by means of transmission spectra obtained with on average 15 atoms trapped inside the cavity (Fig. 1d). The data points and theory curve given in black correspond to the Lorentzian transmission of the empty cavity. In contrast, the transmission spectrum for the two-state atoms coupled to the cavity (red data and dashed curve) displays the characteristic vacuum-Rabi splitting accompanied by a significant drop in the transmission at the empty cavity resonance (probe–cavity detuning $\Delta = 0$). This spectrum is dramatically altered under the conditions of EIT (blue data and theory curve). First, we notice a frequency shift of the vacuum-Rabi resonances due to the 'dressing' of the atom–cavity

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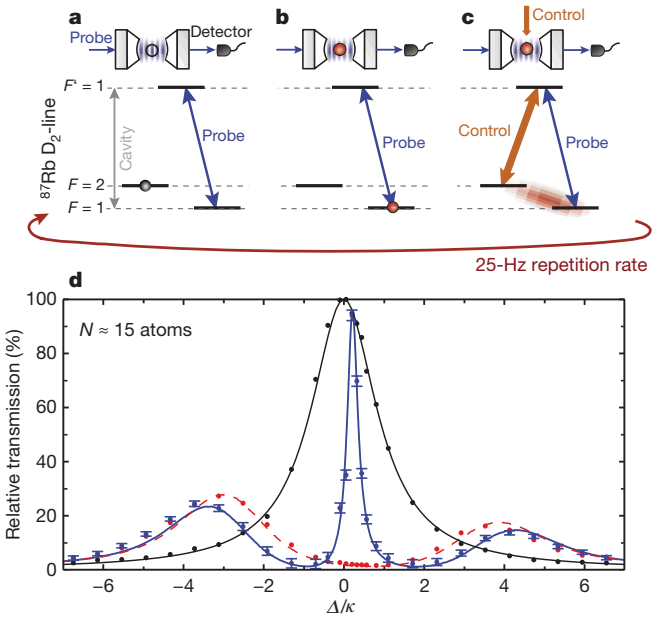

Figure 1 | Experimental protocol and cavity EIT. ^{87}Rb atoms are quasi-permanently trapped inside a high-finesse optical cavity. The cavity is resonant with the atomic $F = 1 \leftrightarrow F' = 1$ transition at 780 nm wavelength. The transmission of the atom–cavity system is probed with a weak laser (probe laser–cavity detuning Δ) for three physical conditions. **a**, With atoms shelved in the hyperfine state $F = 2$, we record the empty cavity transmission as a reference (black data and curve in **d**). **b**, With atoms prepared in $F = 1$, we realize a cavity QED situation and observe a spectrum exhibiting a vacuum-Rabi splitting (red data and curve in **d**). **c**, An additional laser is used to coherently control the optical properties of the atom–cavity system. **d**, Measured transmission spectra for on average 15 atoms coupled to the cavity. A narrow transmission window (linewidth ~900 kHz) observed at the two-photon resonance in the cavity EIT situation (blue data and theory curve) testifies to the existence of a coherent dark state. Experimental parameters: maximum intra-cavity photon number 0.02, control power 9 μW (equivalent Rabi frequency 1.3κ). Error bars shown are ±s.d. and are omitted from the empty cavity and two-level measurement for clarity.

energy levels by the control laser light. Most important, however, is the observation of a narrow transmission window, testifying to the existence of a coherent dark state[27]. We verified that this transmission window appears at the two-photon resonance and can be shifted by changing the control-laser frequency. We emphasize that these results present an implementation of cavity EIT in a completely new regime, where the number of atoms is many orders of magnitude smaller than in previous realizations with atomic ensembles[28,29]. It is the enhanced matter–light coupling per atom inside the optical resonator that allows us to observe EIT with only a few atoms.

The frequency-dependent transmission of our cavity EIT system can be described using a semi-classical theory in the limit of weak probe fields and assuming that almost all the population is in the atomic level $F = 1$. This yields the steady-state transmission T, normalized to the transmission of the empty, resonant cavity, given by:

$$T = \frac{\kappa^2}{|(\Delta + i\kappa) - g^2 N \chi|^2} \qquad (1)$$

Here, N is the number of atoms and $\chi(\delta, \Omega_c)$ the susceptibility of an EIT medium in free space[30] which depends, in particular, on the two-photon detuning δ and the control-laser Rabi frequency, Ω_c. Furthermore, g represents an effective atom–cavity coupling with $g^2 = \Sigma_{i=1}^{N} g_i^2 / N$, where g_i is the atom–cavity coupling of atom i. Equation (1) relates the familiar expression for EIT, $\chi(\delta, \Omega_c)$, to cavity QED with a countable number of atoms. In practice, the transmission spectra are subject to parameter variations and technical noise. Residual atomic motion in the optical dipole trap results mostly in variations of the atom–cavity coupling. Magnetic field fluctuations on short timescales contribute to the ground-state decoherence rate (measured to be about 65 kHz). These effects are incorporated into our theoretical model by averaging over the respective parameter range.

The interest in cavity EIT with a single atom has been put forward in several theoretical works over the past decade, especially in the context of nonlinear optics and quantum information[4–7,14,15]. Its observation is presented in Fig. 2a. The data are an average over 169 complete spectra, each obtained using exactly one trapped atom. For these measurements, the maximum intra-cavity photon number is 0.02. For the case of a two-level atom (red data points), the on-resonance transmission is lowered but the vacuum-Rabi splitting is not resolved. This is a consequence of the atomic motion effectively reducing the coupling to an average value of approximately $0.4g_0$. The weak probe field additionally induces a slow optical pumping to the $F = 2$ hyperfine ground state during the probing period ($t = 50$ μs). This effect has been incorporated by solving the time-dependent master equation for the finite probing interval (red solid curve). The result is an increase in transmission, as compared to the steady state equation (1) (red dashed curve), which is in excellent agreement with the experimental data.

By turning on the control field, we observe cavity EIT with one atom (blue data points and curve). The transmitted spectrum is notably narrowed, and a nearly perfect transparency is obtained. In this respect, the system realizes a quantum-optical transistor with an unprecedented on/off contrast of about 20%, admitting or rejecting the passage of probe photons through the cavity. Beyond that, the power of our approach lies in the preparation of the atom in a dark superposition of two states. This offers the coherent control of the optical properties of a single atom through external parameters. As an example, we investigate the dependence of the transparency window on the control laser power (Fig. 2b, c). The contrast shown in Fig. 2b

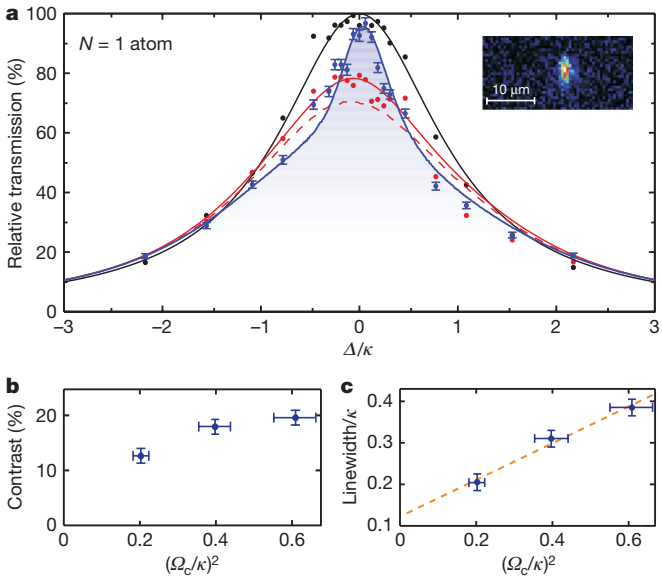

Figure 2 | Cavity EIT with a single atom. Colour coding same as in Fig. 1d. **a**, Measured transmission spectra for exactly one atom coupled to the cavity and a control laser power of 3 μW (equivalent Rabi frequency, 0.78κ). EIT is observed with a maximum transparency of 96% and a measured transmission contrast of 20% with respect to the control laser switched off. The linewidth (see Methods Summary) is ~1.2 MHz. The red solid curve is a solution of the time-dependent master equation for the finite probing interval. The red dashed curve is the prediction of equation (1) for zero control power. Error bars shown are ±s.d. Inset, CCD camera image of a single atom trapped in the cavity (image size, 33 μm × 16 μm). **b**, **c**, The linewidth and contrast of the single-atom transparency feature are tunable by means of the control laser power. The values used are (1, 2, 3) μW and correspond to Rabi frequencies (0.45, 0.63, 0.78) κ. The maximum intra-cavity photon number is 0.02. Error bars shown are ±s.d. for the contrast, ±0.03κ for the linewidth and ±10% for the x axis.

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increases with control-laser power, whereas its maximum value is bound by the difference between empty-cavity transmission and the two-level atom case. We find that the spectral width of the transparency window exhibits linear scaling with Ω_c^2 (Fig. 2c), which is in agreement with equation (1). However, the determination of an absolute value of the EIT linewidth is obstructed by its non-Lorentzian shape due to the overlapping vacuum-Rabi resonances.

Our capability to determine the exact number of atoms in the cavity allows us to investigate the evolution of cavity EIT when adding atoms one-by-one. The experimental results are presented in Fig. 3a–d for respective atom numbers $N = 2, 3, 4, 5$, with all other parameters identical to the $N = 1$ atom case of Fig. 2a. The obtained spectra weakly reveal the vacuum-Rabi splitting, expected to scale with $\sqrt{N}$ for small probe-laser powers. The main effect of increasing the number of atoms is a lower transmission level at vanishing probe detuning ($\Delta = 0$) (red curve). This uncovers the dark-state resonance, while a very high degree of transparency is maintained (solid blue curve). These observations are quantified by evaluating the scaling of transparency, contrast and linewidth as a function of atom number (Fig. 4). The maximum achieved transparency decreases slightly from 96% ($N = 1$) to 78% ($N = 7$), as expected from equation (1). Nevertheless, the measured on/off contrast at the two-photon resonance steadily increases from 20% ($N = 1$) to 60% ($N = 7$), largely owing to the evolution of the two-level spectra. We note that the decrease in the maximum achieved transparency can be compensated by applying a stronger control field, as in the case of Fig. 1d. The dark-state linewidth (see Methods Summary) responds only weakly to the change in atom number (Fig. 4b). The linewidth decreases for $N \geq 3$. In the limit of $Ng^2/(\kappa\gamma) > 1$, here fulfilled for $N \geq 3$, and a negligible ground state decoherence rate, an approximation can be derived from equation (1) for which the linewidth scales as Ω_c^2/N. This is consistent with our observations (dashed curve in Fig. 4b). However, for $N = 1, 2$, the determination of the EIT linewidth is difficult, as explained above. The rise of the two-level atom transmission (used as a reference to obtain the linewidth) compensates the expected broadening, resulting in a nearly constant linewidth for $N = 1, 2$.

Merging the ability to perform experiments with a controlled number of atoms with EIT-based light storage[2] opens up new avenues towards the controlled manipulation of the Fock-state components of propagating light fields. As the number of photons that can be stored in a medium cannot exceed the number of atoms, it now seems

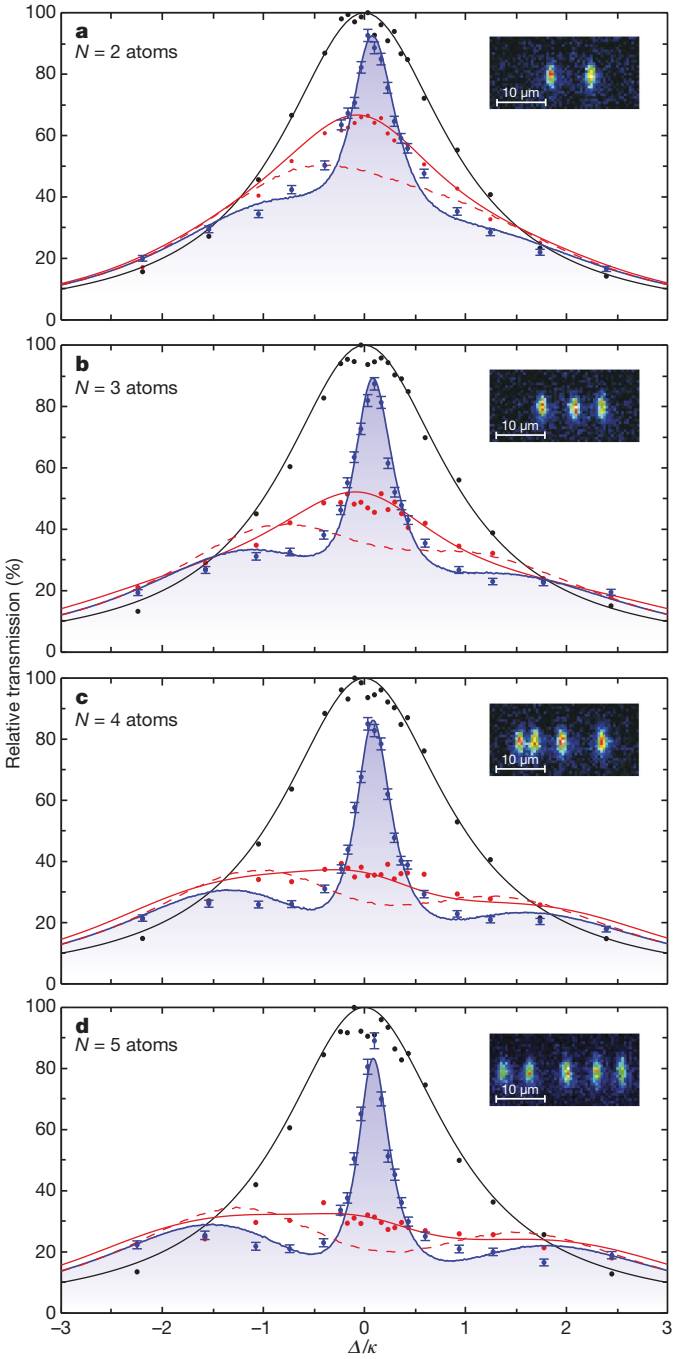

Figure 3 | Cavity EIT spectra for $N = 2$ to 5 atoms. **a**, 2 atoms; **b**, 3 atoms; **c**, 4 atoms; **d**, 5 atoms. Changing the number of atoms enhances the visibility of the dark-state resonance owing to improved contrast with respect to the measurements with two-level atoms coupled to the cavity. The vacuum-Rabi splitting starts being resolved at higher atom number. Colour coding and experimental parameters are as in Fig. 2a. Insets, CCD camera images of the trapped atoms, used to precisely determine their number and physical location in the cavity mode. Error bars shown are ±s.d. and are omitted from the empty cavity and two-level measurement for clarity.

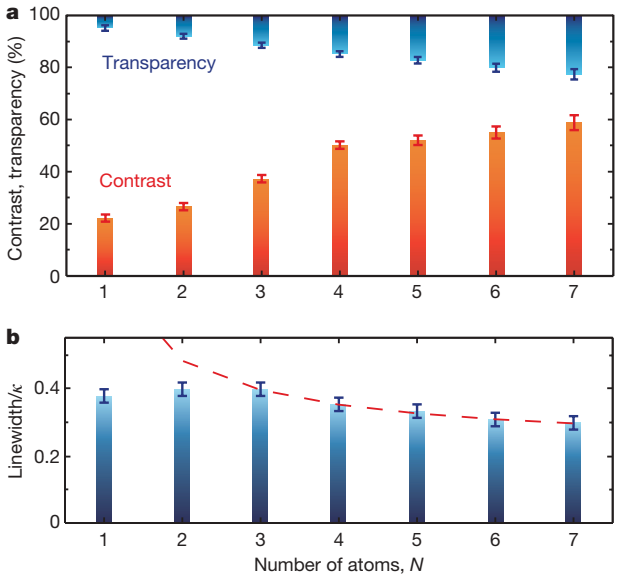

Figure 4 | Measured transparency, contrast and linewidth of cavity EIT with $N = 1$ to 7 atoms. **a**, The maximum transparency (blue bars, top) decreases with the number of atoms from 96% ($N = 1$) to 78% ($N = 7$). Nevertheless, the on/off contrast (red bars, bottom) at the two-photon resonance steadily increases from 21% ($N = 1$) to 60% ($N = 7$) owing to the reduction of transmission with control laser switched off. **b**, For $N \geq 3$, the cavity EIT linewidth decreases with the number of coupled atoms as $1/N$ (guide to the eye, dashed red curve). For $N = 1, 2$, the linewidth is nearly constant owing to the interplay between the increase in the two-level atom transmission and the definition of the linewidth used (see Methods Summary). Error bars shown are ±s.d. for the contrast and transparency and $\pm 0.03\kappa$ for the linewidth.

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possible to write a given number of photons into an atomic register and read out these photons later, while any excess photons would just pass through the cavity. In this way, a highly nonlinear beam splitter could be realized that subtracts a well-defined number of photons from a coherent input field with an initially Poissonian photon-number distribution.

In future experiments we plan to go beyond the coherent manipulation of the average light intensity and investigate the possibility of also controlling the photon statistics. The performance of such experiments would naturally benefit from a larger matter–light coupling constant, g, as can be achieved with a smaller cavity or with better atomic localization. Perfect coupling at an antinode of the cavity mode would result in an increase of g by a factor of 2.5. This would yield a cooperativity parameter, $C = g^2/(2\kappa\gamma) \approx 1$, resulting in a 90% on/off contrast in the single-atom experiment. Larger values of g have already been achieved in other nonlinear single-atom cavity QED experiments[31,32], and would promote the cavity EIT system well into the regime of strong photon–photon interactions. This could bring into reality ideas like that of an EIT-controlled photon blockade in an atomic four-level system, more than a decade after its proposal[10].

During the final stage of manuscript preparation, we learned about related experiments performed in the group of D. Meschede[33].

METHODS SUMMARY

Atom trapping. ^{87}Rb atoms are transferred from a magneto-optical trap into the cavity using a running-wave dipole trap ($\lambda = 1{,}064$ nm). When the atoms reach the cavity, the transfer beam is replaced by a standing-wave configuration, providing strong confinement along its axis. The atoms are cooled using retro-reflected lasers perpendicular to the cavity axis and near-resonant with the $F = 2 \leftrightarrow F' = 3$ and $F = 1 \leftrightarrow F' = 2$ transitions.

Measurement sequence. Empty cavity: following cooling (33 ms), the atoms are pumped to the $F = 2$ ground state using a laser resonant with the $F = 1 \leftrightarrow F' = 2$ transition for 100 μs. Next, a weak probe laser (intra-cavity photon number ~0.02) near-resonant with the $F = 1 \leftrightarrow F' = 1$ transition is applied along the cavity axis for 50 μs. The cavity output is collected using a single-mode optical fibre and directed to a single-photon detection set-up (total detection efficiency ~30%).

Two-level atom: after cooling (3 ms), the atoms are prepared in the $F = 1$ ground state by simultaneously applying pumping beams resonant with the $F = 2 \leftrightarrow F' = 1$ and $F = 2 \leftrightarrow F' = 2$ transitions. The probe beam is applied for 50 μs.

Cavity EIT: after cooling (3 ms), the atoms are again prepared in the $F = 1$ ground state. A control field resonant with the $F = 2 \leftrightarrow F' = 1$ transition perpendicular to the cavity axis is applied simultaneously with the probe. Both lasers maintain a fixed phase relation. During the probe intervals in the last two parts of the sequence, the intensity of the standing-wave trapping field is reduced to minimize a.c.-Stark shift variations of the atomic transitions to below 5 MHz.

Cavity EIT linewidth. We define the linewidth of the resonance from the experimental fit as $|\Delta_{\mathrm{cEIT1}} - \Delta_{\mathrm{cEIT2}}|$, in which $\Delta_{\mathrm{cEIT(1,2)}}$ are the two solutions to the equation:

$$T^{\mathrm{fit}}_{\mathrm{cEIT}}\left(\Delta = \Delta_{\mathrm{cEIT(1,2)}}\right) = \left[T^{\mathrm{fit}}_{\mathrm{cEIT}}(\Delta_{\mathrm{Max\ cEIT}}) + T^{\mathrm{fit}}_{\mathrm{2level}}(\Delta_{\mathrm{Max\ cEIT}})\right]/2$$

where $T^{\mathrm{fit}}_{\mathrm{cEIT}}$ is represented by the solid blue line, $T^{\mathrm{fit}}_{\mathrm{2level}}$ by the dashed red line in Figs 2 and 3, and $\Delta_{\mathrm{Max\ cEIT}}$ is the detuning at which $T^{\mathrm{fit}}_{\mathrm{cEIT}}$ is maximal.

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Author Contributions All authors contributed to the implementation and modelling of the experiment, the interpretation of the results and the writing of the manuscript.

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